

Recovering Proposition 99 Blind: Validating the **natex** Pipeline Against the Abadie–Diamond–Hainmueller Benchmark

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run of record re-verified 2026-07-18, **natex** v0.2.0, seeds 0/1/2 (donor path rng-free)

Abstract

Run blind on the Abadie–Diamond–Hainmueller (ADH) California smoking panel (39 states $\times$ 31 years), the **natex** pipeline reproduces the canonical Proposition 99 benchmark end to end. All three SuDDDS scan methods (greedy, wcc, single_delta; normal model) recover (California, $t_0 = 1989$) exactly — precision = recall = 1.0 over the 31 California records, LLR 13.82; the Bernoulli-model scan is also exact (LLR 13.86), a y-blindness test proves the scan bitwise independent of the outcome, and a Bernoulli($\hat{p}$)-null Monte Carlo ($Q = 99$, seed 1) gives $p = 0.010$ (the dependence-preserving AR(1) null is structurally conservative on a policy dummy, pinning at $p = 1.0$; documented, not claimed as evidence). Deterministic full-pool simplex weights put summed weight 0.955 on ADH’s five published donors, and the synthetic-control ATT over 1989–2000 is -19.514 packs per capita, matching ADH’s canonical average gap of roughly -19 . Nulls are reported as nulls: the exact in-space RMSPE-ratio placebo ranks California 3/39 ($p = 0.077$; ADH’s 1/39 requires their poor-pre-fit exclusion, kept opt-in here), and under the corrected two-sided studentized placebo statistic the source thesis’s claim that all three effect estimates are significant at 5% does *not* survive (ranks 13/39, 7/39, 9/39). By construction this exercise has zero novelty; its value is credibility transfer — external-validity anchoring for the **natex** studies on novel data.

1 Introduction

California’s Proposition 99, passed in November 1988 and effective January 1989, raised the state cigarette excise tax by 25 cents per pack and funded a large anti-smoking campaign. Taxation and media spending each measurably reduced cigarette sales [9], and the program is associated with accelerated declines in consumption and heart-disease mortality [6]. It is also the canonical testbed of the synthetic control method: building on the comparative case-study estimator of [4], Abadie, Diamond, and Hainmueller [2] constructed a “synthetic California” from a weighted combination of donor states and estimated an average gap of roughly -19 packs per capita over 1989–2000. That example anchors a large methodological literature [3, 1, 5] and is arguably the most-replicated benchmark in comparative case studies.

This paper uses it the other way around: not to learn about Proposition 99, but to validate a pipeline. **natex** [11] is an automated natural-experiment discovery-and-estimation toolkit in the LoRD3 lineage [7, 10]; its difference-in-differences arm implements SuDDDS (Subset Discovery of

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Difference-in-Differences) from chapter 6 of [8], whose printed Table 6.1 evaluates the method on this same smoking panel. We run the full pipeline *blind* — the scan is never told the treated unit or the intervention year, and provably never reads the outcome — and ask whether it rediscovers (California, 1989) and reproduces the ADH estimate. It does — exactly, deterministically, and with every deviation from the thesis’s printed values investigated and reconciled. Because the exercise replicates a known benchmark, we write it up short, structurally as validation: its only role is to transfer credibility to the **natex** studies on novel data.

2 Data

The data are the ADH California tobacco panel [2] as shipped in the **natex** dataset registry (dataset **prop99**; 1,209 rows): annual state-level per-capita cigarette sales (packs, from tax revenues) for 39 states, 1970–2000 — California plus ADH’s 38 donor states — with auxiliary covariates (log income, beer consumption, share aged 15–24, retail price). The panel is balanced (39×31 , no missing outcome cells). Treatment is the policy dummy: California from 1989 onward. (The source thesis [8] reports “30 US states” and does not document its panel; ours is ADH’s 39-state pool.)

3 Design and Methods

Blind discovery (SuDDDS). SuDDDS [8], as implemented and corrected in **natex** [11], scans panel records (x_i, t_i, θ_i) — covariates, time, treatment — for a subset of units and an intervention time T_0 at which θ jumps, scoring candidates by a log-likelihood ratio and never reading the outcome y . Protocol, fixed before calibration: windows (5, 8, 10), 4 time bins, 8 restarts, seed 0, and background degree 0 (unit effects only). Degree 0 is required, not tuning: θ is a pure policy dummy whose only time variation is the candidate discontinuity, so any global time polynomial can only fit the jump itself; with degree 1 the leaked slope manufactures a spurious whole-panel optimum (the thesis never reports its background — an unreported-hyperparameter risk). All three scan methods (greedy, wcc, single_delta) are run under the normal model, plus a wcc scan under the Bernoulli model, the corrected default for a binary treatment. A y-blindness test re-runs the scan with the outcome deleted and requires bitwise-identical discoveries.

Calibration. The top discovery’s LLR is calibrated by a fitted-null Monte Carlo with $Q = 99$ replicas and a +1-rank rule (minimum attainable $p = 0.010$), drawing nulls from Bernoulli($\hat{p}$) — the correct null family for a binary θ . As a diagnostic we also run the dependence-preserving AR(1)-unit null, which is *structurally conservative* on a deterministic policy dummy: 38 of 39 units have identically-zero residuals, so the pooled AR(1) fit absorbs California’s step as noise and hands it to every replica; its p pins at 1.0, documented as a regression, never claimed as evidence.

Estimation conditioned on the discovery. Three effect estimators are run on the recovered discovery over the full post period 1989–2000: two-way difference-in-differences (DD), synthetic control (simplex-weighted donor fit on pre-period outcomes, as in [2]), and GESS (greedy control-group expansion, thesis ch. 6). Separately, the **natex** synthetic-control donor path is run on the raw (state, year, cigsale) matrix with the full donor pool and with an $n = 8$ pre-fit-ranked variant; this path is deterministic — no rng anywhere — and bitwise-identical across runs.

Honest inference. Significance is assessed by exact, enumerated placebo tests, with the caveats stated plainly. (i) The in-space RMSPE-ratio placebo [2, 3] refits every one of the 38 control states as

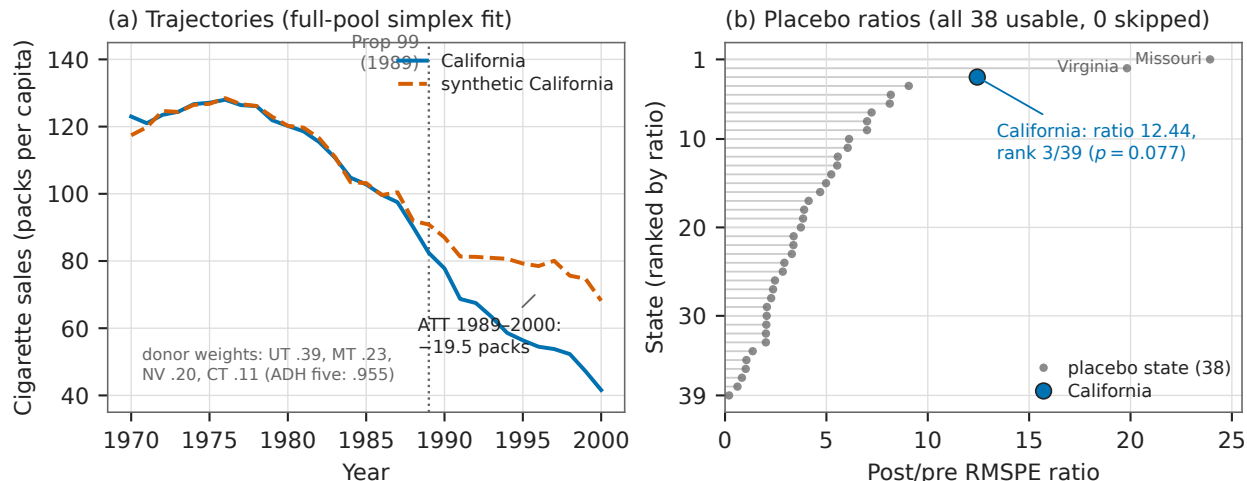


Figure 1: (a) California vs. synthetic California (deterministic full-pool simplex weights), 1970–2000: pre-RMSPE 1.656, post-period gap averaging -19.514 packs per capita. (b) In-space placebo: post/pre RMSPE ratios for all 39 states (all 38 placebos usable, 0 skipped). California ranks 3/39 ($p = 0.077$); the two out-ranking states, Missouri and Virginia, are poor-pre-fit placebos that ADH’s exclusion rule would discard.

a pseudo-treated unit (`exclude_poor_fit = None`: all 38 usable, 0 skipped) and ranks California’s post/pre RMSPE ratio among all 39; ADH’s $p = 1/39$ requires their poor-pre-fit exclusion, kept opt-in in `natex`. (ii) The corrected effect-level test is a two-sided studentized statistic $|\hat{\tau}/se|$ with $+1$ -rank p -values over the 38 enumerated placebo states — under it, genuinely non-parallel placebo states (flat post-shifted gaps with tiny standard errors) can out- t California’s trending gap. (iii) The validation battery’s degenerate cases report “cannot test” (NaN, failed) rather than fabricating passes: composition reduces to a one-row table for a single-unit discovery, and anticipation has zero pre-period treatment variance.

4 Results

Blind discovery is exact. All three SuDDDS scan methods under the normal model return (California, $t_0 = 1989.0$) as the top discovery, with the discovered subset exactly the 31 California records: precision = recall = 1.0, LLR 13.82. The Bernoulli-model scan is also exact, LLR 13.86. The y-blindness test passes bitwise (Table 1, panel A).

The discovery is significant under the correct null. Against Bernoulli($\hat{p}$) nulls ($Q = 99$, seed 1) the observed max LLR 13.86 exceeds the null max 7.95 (null q90 4.67): $p = 0.010$, the $+1$ -rank floor. The AR(1)-unit diagnostic pins at $p = 1.0$, as expected for a deterministic policy dummy (Section 3); recorded as a structural limitation of dependence-preserving nulls, not as evidence against the discovery (Table 1, panel B).

Effects match ADH; the printed thesis numbers reconcile. On the full 1989–2000 post period, DD gives $\hat{\tau} = -27.349$ packs (pre-MSE 51.23), synthetic control gives $\hat{\tau} = -19.514$ (pre-MSE 2.74), and GESS gives $\hat{\tau} = -26.653$ (control: Montana, pre-MSE 18.35). Signs and the pre-fit ordering match the thesis’s Table 6.1 ($-10.94 / -8.96 / -6.67$): synthetic control’s pre-MSE is far

below DD’s, which is exactly ADH’s argument for it. The magnitudes differ by a factor of $\sim 2\text{--}2.5$ because the thesis’s printed values correspond to an unreported effective post window of about five years, while the gap accumulates over time; restricting `natex` to 1989–1993 gives DD -18.8 and synthetic -12.3 , bracketing the printed values (Table 1, panel C).

Donor selection recovers ADH’s pool. The deterministic full-pool simplex fit puts nonzero weight on six donors, with summed weight 0.955 on ADH’s five published donors, and the four heavyweights are exactly ADH’s own {Utah, Montana, Nevada, Connecticut} (weights in Table 1, panel D). $\text{ATT}_{\text{post}} = -19.514$ packs per capita, with pre-RMSPE 1.656 against post-RMSPE > 10 (the ADH-style pre/post contrast); the 8-donor variant gives ATT -22.648 , pre-RMSPE 3.671 (Figure 1a).

Honest inference: what survives and what does not. The exact in-space RMSPE-ratio placebo ranks California 3 of 39 in both variants (treated ratio 12.440 full pool, 6.570 top-8): $p = 3/39 = 0.077$ (Figure 1b). ADH’s 1/39 arises only under their poor-pre-fit exclusion, which `natex` keeps opt-in. The corrected two-sided studentized τ placebos over the 38 enumerated states give DD rank 13/39 ($p = 0.333$), synthetic control 7/39 ($p \leq 8/39 = 0.205$), and GESS 9/39 ($p = 0.231$): the thesis’s claim that all three are significant at 5% does *not* survive the corrected statistic, and we report that as the finding it is (Table 1, panel E).

Run of record and reproducibility. The entire benchmark is executable: running `uv run pytest -m backtest` with `NATEX_DATA` set to the data root, on the modules `test_prop99.py` and `test_prop99_donors.py` under `tests/backtests/`, gives 16 passed, 1 xfailed in 79.74s (re-verified 2026-07-18). The single xfail is a non-blocking instrument-selection stretch goal on a *different* dataset, pinned as a documented finding; all 15 Proposition 99 tests pass. All estimates: `natex` v0.2.0 [11] on the registry copy of the ADH panel (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

5 Caveats and Conclusion

Three caveats bound the claims. First, this exercise has zero novelty by construction: Proposition 99 on the ADH panel is the most-replicated synthetic-control benchmark in economics [2, 1], and a pipeline that failed it would be disqualified rather than informative; its role is external-validity anchoring for the `natex` studies on novel data, and it should be read as methods validation, not as a finding about tobacco policy. Second, effect magnitudes here are $\sim 2\text{--}2.5\times$ the source thesis’s printed Table 6.1 values because `natex` estimates on the full 1989–2000 post period while the thesis’s numbers match an unreported ~ 5 -year effective window; the thesis also never reports its panel (“30 US states” vs. ADH’s 39) or its treatment background. Signs, ordering, and pre-fit ranking agree, and the synthetic-control estimate matches ADH’s canonical ≈ -19 — but the reconciliation must be stated, not glossed. Third, the honest-inference results are genuinely weaker than the textbook telling: without ADH’s poor-pre-fit exclusion the in-space placebo p is 0.077, not $1/39 = 0.026$, and under the corrected two-sided studentized placebo statistic none of the three effect estimators clears 5% on this panel. These are properties of the inference conventions, faithfully surfaced.

Within those bounds, the conclusion is simple: run blind, `natex` rediscovers (California, 1989) exactly by every scan method, calibrates it significantly under the correct null, selects ADH’s donors with 0.955 of the simplex weight, and reproduces the canonical ≈ -19 synthetic-control estimate

Table 1: Proposition 99 backtest: numbers of record (**natex** v0.2.0; seeds 0/1/2, donor path rng-free).

Quantity	Value	Quantity	Value
<i>Panel A: blind discovery (SuDDDS; windows (5,8,10), bins 4, restarts 8, degree 0, seed 0)</i>			
greedy / wcc / single_delta	(California, 1989.0)	precision, recall	1.0, 1.0 (31 CA records)
normal-model LLR	13.82	Bernoulli wcc	exact; LLR 13.86
y-blindness	bitwise identical		
<i>Panel B: discovery calibration ($Q = 99$, +1-rank, seed 1)</i>			
Bernoulli($\hat{p}$) null	$p = 0.010$	observed vs null max	13.86 vs 7.95 (q90 4.67)
AR(1)-unit null	$p = 1.0$, conservative [†]		
<i>Panel C: effects on the discovery, post 1989–2000 (thesis Table 6.1: $-10.94 / -8.96 / -6.67$)</i>			
DD $\hat{\tau}$	-27.349 (pre-MSE 51.23)	synthetic $\hat{\tau}$	-19.514 (pre-MSE 2.74)
GESS $\hat{\tau}$	-26.653 (MT, pre-MSE 18.35)	1989–1993 restriction	DD -18.8 , synthetic -12.3
<i>Panel D: synthetic-control donor path (deterministic, bitwise-reproducible)</i>			
weights	UT .394, MT .232, NV .205	ADH-five summed weight	0.955
	CT .109, NH .045, CO .015	ATT _{post}	-19.514 packs (ADH ≈ -19)
pre- / post-RMSPE	1.656 / > 10	8-donor variant	ATT -22.648 , pre 3.671
<i>Panel E: exact placebo inference (enumerated; no poor-pre-fit exclusion)</i>			
RMSPE ratio (treated)	12.440 full / 6.570 top-8	rank, p	3/39 both; $p = 3/39 = 0.077$
studentized τ : DD	13/39 ($p = 0.333$)	synthetic	7/39 ($p \leq 8/39 = 0.205$)
GESS	9/39 ($p = 0.231$)	thesis “all sig. at 5%”	does <i>not</i> survive

Notes: [†]structural, not evidential: a deterministic policy dummy leaves 38 of 39 units with identically-zero residuals, so the pooled AR(1) null absorbs the step as noise; pinned as a documented regression. Scan p -values use a $Q = 99$ fitted-null Monte Carlo with a +1-rank rule; 0.010 is the minimum attainable. The synthetic-control placebo rank carries one rank of slack because each placebo refits SLSQP weights.

deterministically. The pipeline earns the benefit of the doubt it needs on data where the answer is not known.

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